Problem Set Stop and Wait: Worked Solutions

DataLink – Flow and Error Control lecture, slides 4 and 18–25

Formulas: $T_F = L \div B$ · $T_S = T_F + Prop + Proc_{rx} + T_A + Prop + Proc_{tx}$ · $\mathbf{a = T_S \div T_F}$ · $p = 1 - (1 - p_b)^L$ · optimal timeout gives $T_{Fail} = T_S$
Efficiency: **no errors $1 \div a$** · **ACKs error-free ($q = 0$): $(1 - p) \div a$** · **symmetrical ($p = q$): $(1 - p)^2 \div a$**

Q1. Maximum efficiency

> 21,000-bit frames, 100 Mb/s, 2 km at 200,000 km/s each way, the ACK in a frame of similar length, receiver processing 30 µs, sender processing 60 µs.

1. $T_F = 21{,}000 \div 100{,}000{,}000 =$ **210 µs**. The ACK is the same length, so $T_A = 210$ µs.
2. Prop = 2 km ÷ 200,000 km/s = **10 µs** each way.
3. $T_S = 210$ (frame) + 10 (prop) + 30 (rx proc) + 210 (ACK) + 10 (prop) + 60 (tx proc) = **530 µs**.
4. $a = 530 \div 210 =$ **2.524**.
5. Maximum efficiency (no errors) $= T_F \div T_S = 1 \div a = 0.396$.

≈ 0.396 (39.6%)

Q2. Efficiency with BER 10^{-6} in both directions

1. $p = 1 - (1 - 10^{-6})^{21{,}000} =$ **0.0208**.
2. The ACK frame is the same length with the same BER, so $q = p$: the **symmetrical** case.
3. Efficiency $= (1 - p)^2 \div a = 0.9792^2 \div 2.524 = 0.380$.

≈ 0.380 (38.0%)

Q3. ACK sent on its own ($q = 0$), same a

1. Only the data frame can be in error: $p = 0.0208$ (BER from Q2), $q = 0$.
2. Efficiency $= (1 - p) \div a = 0.9792 \div 2.524 = 0.388$.

≈ 0.388 (38.8%)

> If no bit errors are assumed at all, the answer is $1 \div a = 39.6\%$, the same as Q1. The question builds on Q2, so 38.8% is the intended answer.

Q4. Maximum BER for at least 20% efficiency (Q3 scheme)

1. $(1 - p) \div a \geq 0.2$, so $(1 - p) \geq 0.2 \times 2.524 =$ **0.5048**.
2. $(1 - p_b)^{21{,}000} \geq 0.5048$.
3. $p_b \leq 1 - 0.5048^{1/21{,}000} = 3.26 \times 10^{-5}$.

$p_b \leq 3.26 \times 10^{-5}$

Q5. a = 1.75, 680 B frames, BER 1 in 500,000, q = 0

1. L = 680 × 8 = **5,440 bits**. $p_b = 2 \times 10^{-6}$.
2. $p = 1 - (1 - 2 \times 10^{-6})^{5,440} =$ **0.0108**.
3. Efficiency = (1 − p) ÷ a = 0.9892 ÷ 1.75 = 0.565.

≈ 0.565 (56.5%)

Q6. Maximum BER for at least 40% efficiency (Q5 scheme)

1. (1 − p) ≥ 0.4 × 1.75 = **0.7**.
2. $(1 - p_b)^{5,440} \geq 0.7$.
3. $p_b \leq 1 - 0.7^{1/5,440} = 6.56 \times 10^{-5}$.

$p_b \leq 6.56 \times 10^{-5}$